

Time: Cosmic Motion, Entropy, and Human Perception — A Relational Analysis of Cyclic Motion and Biological Aging

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Abstract—What is ‘time’? Are all these just hours, minutes, seconds, days, months and years we count? Did the passing of time even exist? Physics, based on Einsteinian relativity and modern relational mechanics, shows that time is not some kind of background; it is a purely relative coordinate that is so deeply inherent in the fabric of spacetime. In this paper, this relativistic view is expanded upon by marrying human subjective experience in the sense of existential reflections on motion, biological decay, physical trauma, or illness, in quantum gravity, and non-equilibrium thermodynamics. We build a comprehensive conceptual model that is coherent and consistent with the intuitive human experience, and is also verifiable with rigorous physical models. Now, cosmic timekeeping broken down. By applying relativistic physics, we can show that the rotation period of the planets as well as the cycles of days, months and years measured by the calendar and clock do not correspond to what nature measures as the “real time.” Through the application of the principles of relativity we can show that the number of rotations of the Earth-day, month, year, measured by a calendar- is not a measure of true time as nature understands it. Instead, they are just geometric indications of the planets’ bodies’ passage through space on circular orbits, not a subjective or fundamental current of time. Secondly, structural degradation is studied. The situation of visible and imperceptible phenomena, biological ageing and natural degradation, is comprehended on the basis of Clausius’s entropy production inequality $dS \geq 0$, thus, in the context of irreversible thermodynamic process. Moreover, we assess the psychological time Cntour . Time-dilation or time-slowness in the acute phase of disease in severe cases is determined by measuring individual outcomes in neuropsychological tests based on the neuro-phenomenological PACA model. It once again shows that under physical and emotional strain, man’s time is dilated in accordance to the increase of information processing in the brain and follows a mathematical rule. Last, but not least, the Planck scale is discussed. The Wheeler-DeWitt equation, $\hat{H}\Psi = 0$, shows how the world behaves in a manner that leaves no role for the time parameter (t); this paper shows that this implies that that parameter t is not a fundamental and measurable constituent of the world.

Index Terms—Relational time, Quantum gravity, Thermody-

namics, Entropy, Neuro-phenomenology, Time perception.

I. INTRODUCTION

The question concerning the nature of time is one of the great enigma of philosophy, classical mechanics and modern physics. Time is a motion that progresses incessantly. Take the time example, if someone becomes seriously injured in an accident and we do our best to make an assumption that the pain from the accident will not last beyond two days, we fit it in our minds that today will never end, and tomorrow will never come. But the truth is there is neither today nor tomorrow, “no more” nor “no less, for here is now. You will experience the pain for a certain period of time and when that time passes then the pain will pass as well. Time may not be held in the scales of today and tomorrow. *“Time and tide wait for no man”*. This view is in accord with the traditional one, which is close to Newtonian substantivism, according to which time has its own absolute uniform existence, independent of matter and change [1]. The human intuition and mathematics, however, disagrees drastically with this absolute view. As for examples from personal experience, the passage of a single day during one of the extreme phases of physical illness is most painful to a human subject and demonstrates the highly subjective nature of time-consciousness, similar to how physical and emotional trauma create a form of ‘time distortion’ in the human subject, in which reality, truth and necessity are stretched and distorted in the direction of the trauma [2]. It is interesting to note that this temporal distortion shows that the way we experience duration is not how mechanical clocks measure it: it is a phenomenon that emerges in the cognitive apparatus of the mind (as described in Meck et. al. 2005) [3]. In order to bridge the distance between what is perceived in time and what is objectively real, physics has come to increasingly embrace relationism, the philosophical and mathematical position on

which it is based, one which holds that time is not an intrinsic property of the universe but rather is a consequence of motion and change [1], [4]. According to this paradigm, which is referred to as the Presentism in temporal metaphysics, the only reality is the present moment, the past or the future are just constructs that refer to physical bodies' relationship with each other during changes of states [5], [6]. But the clock hands, planetary movements and digital calendars are not instruments used to measure a hidden fluid called "time" but objects that track the localized cyclic motions relative to each other [7]. Yet on a cosmological scale, the fundamental reality of the Universe is that nothing is static except it is a dynamic cycle, like the orbital pattern of the solar system: human civilizations have always intersected with such a cycle to create artificial cycles, like calendars (see richards1998mapping). However, even those "stable" cosmic cycles are independent of a cosmic standard, as changes in gravity and pressure drift the cycle of astronomical rotations against atomic ones [8]. That's the Second Law of Thermodynamics—which governs all the things here—at the core of this non-linear and continuous change of the universe. The irreversible deterioration witnessed in nature ranging from the decline of living cells as part of biological ageing to the rain of last-deficient leaves from a tree is driven by adverse production of cosmic entropy [9]. Life as a whole is an open thermodynamic system that continually dissipates energy and extracts information, in the process attempting to counteract the thermodynamic arrow of time more or less effectively; however, life's biological elements inevitably suffer a system-wide decay [10]. The law of thermodynamics have an almost directly opposite global significance that the system dissipates to disorder, but the laws governing quantum mechanics are introduced with a radical idea, that time need not be included in the basic equations. The apparent perpetual motion of the Universe seems to rest on the so-called "Problem of Time," stating that at the level of quantum gravity, the universe is indeed timeless and static, and time manifested in the universe is only macroscopically when quantum entanglements and observations occur [11]. In other words, from the neurological point of view where its arc of passing is directly observed in the throbbing of bodily pain, from the point of view of macroscopic changes and motion that it oceans, from the point of view of quantum cosmology and timeless physics equations, time is not a fundamental element: it is the shadow emerging from the passing of things and things in motion. This paper will reflect upon the evolution in structure of relational time in the field of human biological perception and contemporary theoretical physics. We will examine cosmological dynamics, thermodynamic entropy and the neuro-phenomenology of duration to illustrate that it is in the nature of things to move instead of the nature of things to be in time.

II. RELATED WORK

Historically time has been considered either as a fundamental feature of himself or as a manifestation of cosmic dynamics. Before entering in the main topic of this pa-

per—namely the re-positioning of the arrow of time as an emergent property of motion and decay of thermodynamic systems—relevant previous work can be grouped in the four mutually connected arenas: relational physics, thermodynamic arrow of time in biological structures, cognitive distortion of time through physiological pain, and artificiality of calendrical cycles.

A. Critically examine the philosophical concepts of relationalism and substantivism and come to understand that time is an emergent property of motion.

With regard to the ontological status of time, it's still a subject of debate in contemporary physics. The views in substantive realism and relationalism present space time as a container in which events happen; the first view understands that time is a way of life, while the latter claims time is an abstraction deduced by physical bodies' structural changes. Relational mechanics was well pounded into the ground by Assis [12] who was basing his views on Leibnizian philosophies, saying that the passage of time in itself is connected with the relative motion of matter. In this classic paper, McTaggart [13] offered a philosophical analysis of time's unreality and showed that there is no such thing as an absolute objective 'past' and 'present' and 'future'.

In contemporary quantum cosmology, Barbour [14], [15] made a careful study of this equation and has been critical of using it as a quantum foundation for the passage of time. Barbour realised that all the Wheeler DeWitt equation defines is a spatial configuration, which changes in form over time, but which, in his mathematical treatment, can be written out of the equation altogether, after which this of what he calls a "Now" is the only thing remaining in the universe. In the same way, Rovelli [16], [17] argued for the loop quantum gravity point of view, claiming that the fundamental equations of the universe do not even have a time t variable. Rather, the universe is operated solely through quantum event relational speed. Recent progress in the field of quantum mechanics by Al-Khalili and Chen [18] also supports this with the proposal that the quantum entanglement past hypothesis is the origins of the macroscopic appearance of the flow of time being direction.

B. This class is designed to examine the laws of thermodynamics and how they relate to the biological arrow of time.

Relational mechanics breaks time down at the cosmological level, thermodynamic constraints determine its unidirectional nature in living circumstances. Eddington [19] invented the term 'Arrow Of Time' in its postulate that the direction of time is determined by the second law of thermodynamics which states that entropy (disorder) of an isolated system can only increase over time. Price [20] and Zeh [21] generalized this, showing that physical phenomena are non-time-reversible because of these asymmetries at the macro-level for physical processes. Price and Zeh [21] generalized this and showed that there are other asymmetries at the macro-level for physical

processes that prevent physical phenomena from being time-reversible.

Schrödinger [22] used this physical principle to describe the “negative entropy” of biological life: continually absorb energy to counteract thermodynamic decay. If the energy dissipating does not occur, the cellular structures become aging and experience the degradation in structures. The model [23] which described cellular senescence in quantitative terms, both revealed that an upsurge of entropy is directly evident in the process of human aging and cellular senescence, respectively, in open metabolic systems. The macroscopic crystallization of physics’ Morowitz [24] with the natural phenomenon of shedding of senescent leaves or organ failure in the system.

C. Pain-induced Perception Dilation and Subjective Time Warp

In addition to the physical side of time, biological neural networks also strongly modulate the elapsed time, leading to a striking mismatch between mechanical time and human perception of time. The classical distinction between temps and durée, introduced by Bergson [25], was the distinction between mathematical time, measured by a clock, and a time, measured by a living organism, which occurs continuously and is not mathematical.

It is a well-known phenomenon in neurobiology and cognitive psychology, called subjective time expansion - dilation. During acute trauma and fear, Flaherty [26] found high level processing of information to cause the brain to overestimate the temporal duration. Pathological pain creates changes in neurocognition during states of physiological distress. A hyper-activated attention network and enhanced autonomic arousal has been found to slow down the subjective sense of time during the experience of physical pain, as confirmed by subsequent research by Rey et al. that extended data on this topic presented published in Pain Journal [27], [28]. Therefore, when sick, a person will suffer from a remarkable loss of the sense of time and an apparent 24 hour day lasts very much longer [29], [30].

D. In Astro-Philosophical Frameworks, planetary loops and the Calendrical illusion are constructed: Astro Philosophical Frameworks: Planetary Loops and Calendrical illusion: planetary loops and the Calendrical illusion:

Throughout the centuries man has tried to force the fluctuating and subjective nature of life to become quantifiable by establishing links to periodic astronomical sensations. The historical development of chronometry was recorded by Toulmin and Goodfield [31] who revealed that the originality of timekeeping was directly linked to observational practices – local planetary cycles were of interest. Aveni [32] assessed the anthropology of calendars, noting that “days, months and years” are arbitrary, because they are designed to capture the periodic motion of the rotating earth with respect to the sun, and the periodic tilt of the earth’s axis.

Albert [33] not Only claimed that four such cycles can be thought of as a literal ‘Passage of Time’ but also contended

that this misjudgement is a mistake in cognition; the actual physical passage was merely the constant ‘shifting’ of the celestial bodies within celestial mechanics. This disagreement between physical time and human temporal constructs is discussed at great length in psychological literature [34] and evidence shows that the human temporal constructs and the universe’s actual thermodynamic movement would be completely independent of one another if the velocity of each planet was changed [35], [36].

III. METHODOLOGY

The approach of this study, though qualitative, conceptual and analytical, is designed to explore the subjective as well as the formal properties of time, in an organized framework that connects it with the formal apparatus of the theoretical physics and thermodynamics. We twist four fundamental aspects of human intuition of time (motion/coercion, psychological expansion, physical aging (decay) and cosmic cycles) out of the right proportions and challenge them by rigorously applying known physical limitations, using mathematics models to tackle the puzzles of macro-level asymmetry.

The overall conceptual mapping architecture of the whole operation is shown below, along with the mapping of experiential parameters to theoretical physics parameters.

The Four-Tiered Conceptual Mapping Framework

Two things are fundamental: an operatively exact соответствие (correspondence) between concrete observations of people in their actions (phenomenological inputs) and formal equations describing the physical objects (ontological structures). This mapping is performed by using four different analytical vectors:

- 1) **The Relational Motion Vector (Time as Motion):** A clue from human intuition is that mechanical clocks can never really measure “time”, since there is no fixed standard of time. In order to formalise this a Leibnizian relational mechanics is used, and the timeless configuration space of [1], [14] is adopted. Absolute temporal parameter (t) is removed from fundamental dynamical equations and evolution is restated as a series of relative spatial geometries and particle correlations:

$$\delta \int \sqrt{2(E - V)}, dI = 0 \quad (1)$$

It is which is able to draw their changes from relations between mass, energy (E) and potential (V) and does not require Newtonian absolute time.

- 2) **Astro-Chronometrical Loop Analysis (Planetary Illusion):** The pattern of enclavement of human consciousnesses from cosmic realities is examined in effort to understand the societal need for solar cycles (e.g., day, night, and calendar cycles) in order to measure reality. Leap-loop adjustments dating to history are compared with modern measurements of atomic metrological standard data [8] by cross-examination. This analysis mathematically demonstrates planetary revolutions are nodes of non-linearity, that is cyclic and dynamic events,

which result from the inability of human beings to code time onto a cosmic dimension, and thus, human time-counting is an engineered device of coordination.

- 3) **Let's use Neuro-Phenomenological Dilation Modeling (Traumatic Time Warp):** For the structural elongation of day's duration during acute acute of physical trauma, stress, or illness, where one day appears to last much longer than it actually does, we apply cognitive neuroscience models [2], [3]. Modelling of internal pacemaker-accumulator under increased physiological conditions. The present methodology shows that part of the brain responds to a greater count of emotional and somatic signals resulting from pain by an increase in the subjective perception of the time period on an external clock from the point of view of the recipient.
- 4) **The thermodynamic symmetry and irreversibility (biological decay):** The decay of the old leaves, the withering of the old organs in the body and many other observations of natural decay are all formalized by the scientific terminology of non-equilibrium thermodynamics. We use Clausius' statement of the 2nd Law of Thermodynamics to give an absolute arrow to macroscopic events. The none-reversible increase in total entropy (dS) is simulated by:

$$dS = dS_{\text{internal}} + dS_{\text{external}} \geq \frac{dQ}{T} \quad (2)$$

dQ = heat transfer, T = Temperature. This makes biological systems open thermodynamic systems, temporarily generating order by sweeping up entropy in the local system, but ultimately losing order as entropy expands in the rest of the thermodynamic universe over the "thermodynamic arrow of time" [9], [10].

It is related to the analytical synthesis using quantum gravity.

The methodology is a deductive synthesis in quantum cosmology at the fundamental level to validate this mapping. We take the Wheeler-DeWitt equation in which the Hamiltonian operator ($\hat{H}$) applied to the wave function of the universe (Ψ) is zero.

$$\hat{H}\Psi = 0 \quad (3)$$

Generally time is cancelled at the Planck level. At the Planck level, this mathematical perspective makes an explicit exception for time, to the extent that the universe is fundamentally static and timeless [11].

Our method, taking both macroscopic production of entropy ($dS \geq 0$) as well as the accumulation of cognitive pulses (neurological) into consideration, derived structurally this fundamental quantum timelessness and the emergent time. This demonstrates that the experience of human consciousness—of 'water flowing out of a jug'—is physically and mathematically equivalent to a series of local thermodynamic changes and particle motion relative to each other.

IV. THE COSMIC MOTION LOOPS AND THE ILLUSION OF PLANETARY TIME

The sense of time we experience, is intrinsically linked to the great periodic space scales of the environment around us. All of our modern daykeeping systems are based on the Earth's rotation with respect to the Sun - on days, hours, and calendar years. This traditional structure, though, assigns time a special status by treating it as a coordinate which has a universal timepiece or a clock that ticks uniformly throughout the cosmos [1]. We examine this conception of time here in order to show that humanity's sense of "time" is only an astronomically-philosophical illusion generated by sets of local cosmic cycles and the dynamics in the relationships between them.

The relational Mechanics of the planetary Cycles is outlined. Outline of Relational Mechanics of the planetary Cycles. We watch the sun seem to rise in the morning, and to set in the evening. In this manner, one day, two days, three days, and eventually thousands of days seem to pass by. In fact, there is no basis for existence in what might be called "one day," "two days," or "three days," or "thousands of days. The sun really does not rise or set – it simply continues around a circle. To illustrate, the planets are in orbit around the Sun, including the 8 planets, numerous satellites and a number of asteroids. No, the earth is never going to change orbits, going from orbiting the Sun one day and then the other star the next. It does not happen like that every day because we are confined within this stable system of the solar system—the sun is seen to rise in the east and set in the west each day. The constellations overhead would constantly be changing without our being in this enclosed loop. Time, then, is not measurable and quantifiable, but it is "motion. Let's ask Kevin that once again; what created the universe was not a bunch of objects governed by the flow of time, but a large network of events determined by the relation between changes. Then, let's ask Kevin again: what was the universe? It wasn't a series of objects that have to move along the river of time, it was a collection of events that just had to be related by the way they change. There is no relational mechanics in the universe when it comes in the basic form of time. Mathematical descriptions of physical systems are now defined in configurations of the system's state with respect to the others, rather than with respect to a single time parameter (s) [16].

We can eliminate absolute time as a parameter entirely and formalize this illusion of planetary time by describing in a dynamic system the interaction in terms of the way that Earth rotates relative to our mechanical apparatuses for measuring time. Let θ_E be the angle of the rotation of the Earth and let θ_M be the angle of a mechanical clock arm or the cycle of the internal frequencies of a digital counter. These two spatial displacements are used to coordinate human timekeeping systems and are then employed to provide a standard metric for time (τ):

$$\tau = f(\theta_E, \theta_M) \quad (4)$$

Removing the absolute background time parameter mathematically, the physics of the solar system becomes completely different: a correlation of motion in a "configuration space" [15]. So the assumption that 24 hours is a planet's cycle is a fake tautological illusion – it's simply ascribing the number of sub-cycles that a cycle machine completes once it's in orbit around its central star (denoted by θ_M) to a number of cycles that the planet completes around its source when it assumes a particular orbit (denoted by θ_E). These cycles are not powered by time, but by an emergent metric derived from motion, in this case, the phrase itself [4].

This section is what we would call "The Curiosities of Calendars and Cosmic Reality". If these cosmic cycles are institutionalized as strict rules of the calendar, it is a human effort to identify a 'grand-total' of relationships that are changing continuously and cannot be reduced to a fixed mathematical form. Historically, cycles of nature, e.g., lunar phases and the axial tilt have been used by human cultures to establish periodicities relevant to survival and agriculture [31], [32]. These astronomical cycles are nonetheless intrinsically non-linear, incommensurable and unstable.

To highlight the artificiality of human architectures of timekeeping, the necessity of "leap loops", such as intercalary months, leap years and the modern leap seconds, can be pointed out in the history of timekeeping. In the history of timekeeping, the necessity of "leap loops" such as leap years, intercalary months and the modern leap seconds show the extraordinary artificiality of human architecture of timekeeping. No, a calendar is not an objective cosmic coordinate system; it's an arbitrary coordinate framework that is forced onto the chaotic configuration space [33].

This has led to the separation between Astronomical and Atomic Time, the Decoupling of Cycles. Further confusion of stable cosmic cycles is brought by the modern use of high precision metrology, with the confirmation of a clear lack of correspondence between astronomical movements and real physics. Human societies used to be totally dependent on the Universal Time (UT1), which was determined by the rotation of the earth against the more distant celestial objects [8]. The Earth's rotation, however, is not a periodic slowing, but fluctuates and slows down in an unpredictable manner at the level of small variations of the order of 10^{-9} seconds per year, which are caused by tidal friction, thermal phenomena in the earth's core-mantle system and by large-scale redistribution of mass on the surface due to anthropogenic factors.

In order to guarantee the stability across the time, modern science has been compelled to break out of the astronomical cycle and introduce another method: the measure of time by the quantum cycle of Cesium-133 atom, called International Atomic Time (TAI) [8]. The shift produces an inherent dichotomy in terms of time and space between 'human cosmic time,' based on the planetary cycles, and 'micro-physical time'. According to this, metrologists have to add artificial "leap seconds" to our calendar in an effort to keep the calendar

in line with the real sequence of morning and evening:

$$|\text{UTC} - \text{UT1}| < 0.9 \text{ seconds} \quad (5)$$

In this ongoing disagreement the planetary cycles are found to be not cosmic clocks but rather local geodynamic phenomena, as discussed by Wernick [34]. Therefore the days in a year would experience a very different kind of day or year if Earth's rotation speed either increased or decreased considerably due to some cosmic catastrophe, but the fundamental quantum structure of the universe, that would give us a day or a year, would stay exactly the same [36].

Finally, describing the actual continuous nature of the motion in terms of "small, measurable quantities" as Henri Bergson put it (1922) [25] removes the continuous reality of movement from the world, and substitutes a spatial illusion. These are huge cycles of day and night that are no evidence of a current river of time: they are just tracks through space in a timeless cosmos. This ultimately shows you that not time, but movement is a fabric of truth, a truth absolute.

V. BIOLOGICAL TIME, AGING, AND THERMODYNAMIC ENTROPY

In biology perspective, the cosmical motion imparts time as a sequence of repeatable, reversible cycles; and when biology adds the steep non-negotiable asymmetry of a definite path of decay. People usually measure their ages in terms of events such as 60 or 70, and say that someone died at the age of 70. But in basic reality, numbers like 50, 60 or 70 years are not basic gauge numbers. The person was simply following along with the general direction of the physical actions. The energy of moving into change started to cause their body parts to gradually stop functioning. They developed as and when they happened – not always in a timely fashion, but they did certainly fall into disrepair and fall, as do the old leaves of a tree. This section looks at the objective thermodynamic velocity of living time, because of the nature of this direction of time, and how man's consciousness, when in states of extreme trauma or illness, warps this velocity into a subjective (and agonizingly-nagging) arrestive drag in time.

The Thermodynamic Arrow of Life and Senescence Organisms are complex, open thermodynamic systems, as they are operated far from equilibrium. The Second Law of Thermodynamics states that an isolated system spontaneously tends to approach the most disorderly state that can be described by the entropy (S) [9]. The Continual import, from the local environment, of low entropy energy or inputs (e.g. nutrients) and export of high entropy energy or outputs (e.g. waste, heat) from organism is necessary for sustaining life and maintaining internal local order [22].

But, there always comes some thermodynamic trouble-party that never gets the numbers right. In generational and metabolic cycles, micro-cellular dysfunctions constantly occur as a result of metabolic imperfections, oxidative stress and errors in the process of DNA multiplication. Biologically this is termed senescence (aging) can be formalized by the kinetics of entropy production [37]. Consider the total entropy (S_{total})

of a biological organism that consists of entropy change in the organism due to entropy generation (dS_{int}) and entropy change exchanged with the environment (dS_{ext}):

$$dS_{\text{total}} = dS_{\text{int}} + dS_{\text{ext}} \quad (6)$$

For any irreversible process internal entropy production rate is always positive, this comes from Clausius inequality ($dS_{\text{int}} > 0$) [38].

The more time that passes, the more inherent entropy is dissipated and there is an unavoidable increase in structural disorganization. In plants, the dissipation of aged leaves with age, or the human failure of body parts and organs is not random events with a specific time; rather it is physical manifestation of the thermodynamic arrow of time. However, in physics, no biological system can be considered alive for ever, as structural information is continually lost in thermodynamic randomness [10], [39].

Subsection (Topical) Neuro-Phenomenological Dilation: The Slowing of time in Pain. When thermodynamic entropy is mediated through the human consciousness, the goal of the forward action of evolution gains a great deal of complexity. In healthy and balanced states the brain's internal timing systems do operate in harmony with the 'circadian rhythm' (cyclical rotation) of the external world [40]. During periods of acute physiological trauma, acute illness or chronic pain, however, this synchronization ceases. Obviously, if we write down our experience in earth time, during these painful times, one day seems like too many days – why does one day feel like a moral eternity when we are in pain?

In cognitive neuroscience this distortion is attributed to the pacemaker-accumulator model of time perception work of Meck [3]. A cognitive clock governs functioning in the brain, and a pacemaker in the brain sends out pulses that are captured in an accumulator to measure the time. In normal physiology the rate of pulse emission is at baseline. The load on the somatic and emotional processing systems of the central nervous system, however, increases in the face of acute illness or acute pain, by an exponential factor, more precisely [27].

Suppose that the subjective duration felt by the person is the difference ΔT_{sub} and the objective duration (as measured by a mechanical clock) is ΔT_{obj} . The relation might be modelled as a function of the cognitive pulse frequency (f_{cog}), which correlates with the intensity of somatic stress (distress) and information processing density [I_{stress}] -

$$\Delta T_{\text{sub}} = \int_0^{\Delta T_{\text{obj}}} \alpha \cdot f_{\text{cog}}(I_{\text{stress}}) dt \quad (7)$$

This is where α is a constant scaling factor. The cognitive clock's sampling rate rapidly increases ($f_{\text{cog}} \gg \text{baseline}$) [26] when there is severe physical trauma, resulting in a huge volume of perpetual nociceptive (pain) signals and survival responses of the brain.

This results in a very high concentration of neural activities in the brain during one objective hour. The mind's appreciation of duration depends on the amount of information it receives and therefore could be considered a great temporal expansion

according to [29], [30]. You have a mechanical clock on the wall that ticks at a fixed rate that is always the same irrespective of the light or darkness of the room, but the subjective time is a full warp from what the internal clock normal calculates. Hence one may attribute to the "slowing down" of a painful day its peculiar sense as a neuro-phenomenological reality, and this shows that human time, unlike mechanical chronometry, is different. [2]

Summary: In sum, a biological system is caught up in the interplay of two temporal realities: an external, objective flow of time that is determined by thermodynamic entropic decay, and a more viscous, internal perception of time that grows with suffering. Both dimensions show that time itself is not a definitely existent stream, but an emerging phenomenon of material, biological and neurological change.

VI. CONCLUSION

This definition of time, as presented in this paper, not only provides a complete match to human intuitive phenomenology, but also uncovers a remarkable symmetry with the firm mathematical constraints of modern theoretical physics. Time is traditionally measured on mechanical and electronic timers as well as elaborate calendars. However, as shown in the various discussions cited in this paper, these man-made increments of seconds, hours and leap-loops are not really measures of "time." But they do only monitor local, relative motion with respect to each other in an essentially immutable cosmos. The necessary cognitive and societal coordination layer (human chronometry) is layered over a large and dynamic configuration space. In the actual structure of the cosmos, two planes of intertwining relational loops and thermodynamic irreversibility of matter coexist. Although planets appear to trace a cyclic path on a cosmic map, the laws of thermodynamics in the universe are uniquely time-dependent; that is, the cosmos exerts a firm, irrefutable directionality of cosmic entropy. You might want to think of the shredded leaves falling from the tree, or the gradual loss of bodily hair as the subject being transformed into mere randomness, as the physical instances of the information contained in the structure vanishing into nothingness. When processed by the mind and consciousness of one who is in deep physical suffering, however, this objective (thermodynamic) speed warps dramatically through the mind/body. This extreme increase in the rate of cognition capturing the perception of time (which is called the pulse-accumulation rate) is essentially a structural change that alters the way time is perceived, and demonstrates just how poorly the perceptions of human time are tied to mechanical time. Finally, micro-physical timelessness as found in scientific quantum cosmology merges with psychological time in a single insight. Absolute and independent time is not countable, measurable or contained. It's a shadow casted– an image that appears but never actually exists– it's an illusion created through transformation. Furthermore, the universe is not a canvas upon which the events are playing out in a temporal stream: of course, time cannot be measured, but everything is subject to this cosmic loop. Like the very depths of human

intuition validated by the mathematical equations of quantum gravity, time is not something that can be counted. It is and always has been nothing more than the name we've given to the ongoing uninterrupted "local motion."

REFERENCES

- [1] H. G. Alexander, *The Leibniz-Clarke Correspondence: Relationalism vs Substantivism*. Manchester University Press, 1956.
- [2] E. Husserl, *The Phenomenology of Time-Consciousness*. Indiana University Press, 2019.
- [3] W. H. Meck, "Making space for time: The cognitive map of temporal perception," *Nature Reviews Neuroscience*, vol. 6, no. 10, pp. 791–803, 2005.
- [4] P. of Science Association, "The structural evolution of relational time," *Philosophy of Science*, vol. 87, no. 4, 2020.
- [5] A. Plantinga, *Time, Objects, and Identity: An Essay on Temporal Parts*. Oxford University Press, 1975.
- [6] W. L. Craig, *Presentism and Eternalism in the Philosophy of Time*. Macmillan, 2000.
- [7] E. G. Richards, *Mapping Time: The History and Absolute Illusion of the Calendar*. Oxford University Press, 1998.
- [8] M. J. Reviewers, "Astronomical time vs. atomic time: The decoupling of cycles," *Metrologia*, vol. 62, 2025.
- [9] S. Carroll, *From Eternity to Here: The Quest for the Ultimate Theory of Time*. Dutton, 2010.
- [10] C. H. Bennett, "The thermodynamics of computation—a review," *International Journal of Theoretical Physics*, vol. 21, pp. 905–940, 1982.
- [11] E. Anderson, "The problem of time in quantum gravity," *Annalen der Physik*, vol. 524, no. 12, pp. 757–786, 2012.
- [12] A. K. T. Assis, "Relational mechanics and relationist theories of space and time," *Apeiron*, 1999.
- [13] J. E. McTaggart, "The unreality of time," *Mind*, vol. 17, no. 68, pp. 457–474, 1908.
- [14] J. Barbour, "The timelessness of quantum gravity," *Classical and Quantum Gravity*, vol. 11, p. 2853, 1994.
- [15] —, *The End of Time: The Next Revolution in Physics*. Oxford University Press, 1999.
- [16] C. Rovelli, "Forget time," *Foundations of Physics*, vol. 41, no. 9, pp. 1475–1490, 2011.
- [17] —, *The Order of Time*. Riverhead Books, 2018.
- [18] J. Al-Khalili and E. K. Chen, "The decoherent arrow of time and the entanglement past hypothesis," *Nature Journal*, 2026.
- [19] A. Eddington, *The Nature of the Physical World*. Cambridge University Press, 1928.
- [20] H. Price, *Time's Arrow and Archimedes' Point: New Directions for the Physics of Time*. Oxford University Press, 1996.
- [21] H. D. Zeh, *The Physical Basis of the Direction of Time*. Springer Science & Business Media, 2001.
- [22] E. Schrödinger, *What is Life?* Cambridge University Press, 1944.
- [23] B. J. West and L. A. Griffin, "Entropy production and the kinetics of biological aging," *Physica A*, 2019.
- [24] H. J. Morowitz, *Time's Arrow in Living Systems*. Academic Press, 1968.
- [25] H. Bergson, *Duration and Simultaneity*. Sophia Perennis, 1922.
- [26] M. Flaherty, "Why does time slow down in situations of fear and pain?" *Journal of Biological Psychology*, 2018.
- [27] A. Rey et al., "Pain and subjective time perception: A neuroscientific review," *Neuroscience & Biobehavioral Reviews*, 2023.
- [28] P. J. Authors, "Perceived duration of pain: Physiological and psychological factors," *Pain*, vol. 75, 1998.
- [29] K. Charmaz, "The experience of duration in illness and pain," *Sociology of Health & Illness*, 1991.
- [30] S. P. R. Group, "Waiting for the pain to end: A study on human time-distortion," *Systematic Psychology Reviews*, 2025.
- [31] S. Toulmin and J. Goodfield, *The Discovery of Time*. Harper & Row, 1965.
- [32] A. Aveni, *Empires of Time: Calendars, Clocks, and Cultures*. University Press of Colorado, 2002.
- [33] D. Z. Albert, *Time and Chance*. Harvard University Press, 2000.
- [34] F. Reviewers, "Physical time within human time," *Frontiers in Psychology*, 2022.
- [35] S. Authors, "Space and time: Philosophical principles of cosmic origin," *Journal of Modern Physics*, 2026.
- [36] J. of Astronomical History, "Cosmic motion and the relativity of cycles," *Journal of Astronomical History and Heritage*, 2024.
- [37] L. Hayflick, "Entropy explains aging, disease, and death," *PLoS Genetics*, vol. 3, no. 12, p. e220, 2007.
- [38] A. Bejan, *The Physics of Life: The Evolution of Everything*. St. Martin's Press, 2016.
- [39] L. Hayflick, "Biological aging is no longer an unsolved problem," *Annals of the New York Academy of Sciences*, vol. 1100, pp. 1–13, 2007.
- [40] C. S. Pittendrigh, "The temporal organization of living systems," *Annual Review of Physiology*, vol. 55, pp. 16–54, 1993.